

A Distributed Two-Level PMU-Only Linear State Estimator

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Linear State Estimator

– State Estimation Problem (WLS)

- Complex Number Form:

$$\text{Min } \tilde{\mathbf{r}}^T \tilde{\mathbf{W}} \tilde{\mathbf{r}}$$

$$\text{s.t. } \tilde{\mathbf{z}} = \tilde{\mathbf{H}} \tilde{\mathbf{x}} + \tilde{\mathbf{r}}$$

- Real Number Form:

$$\text{Min } \mathbf{r}^T \mathbf{W} \mathbf{r}$$

$$\text{s.t. } \mathbf{z} = \mathbf{H} \mathbf{x} + \mathbf{r}$$

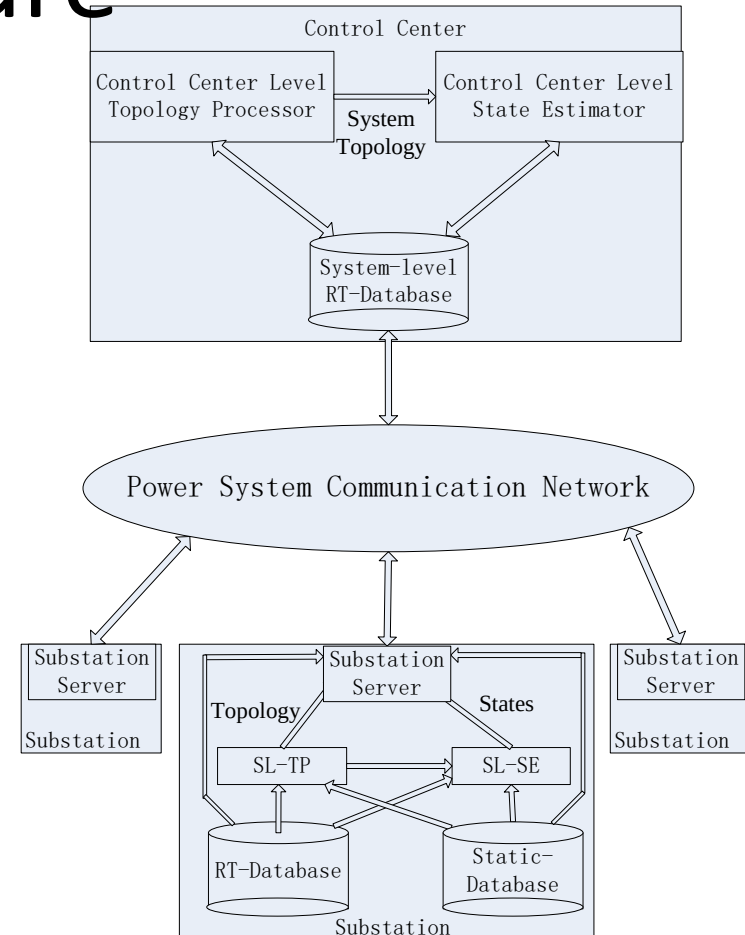
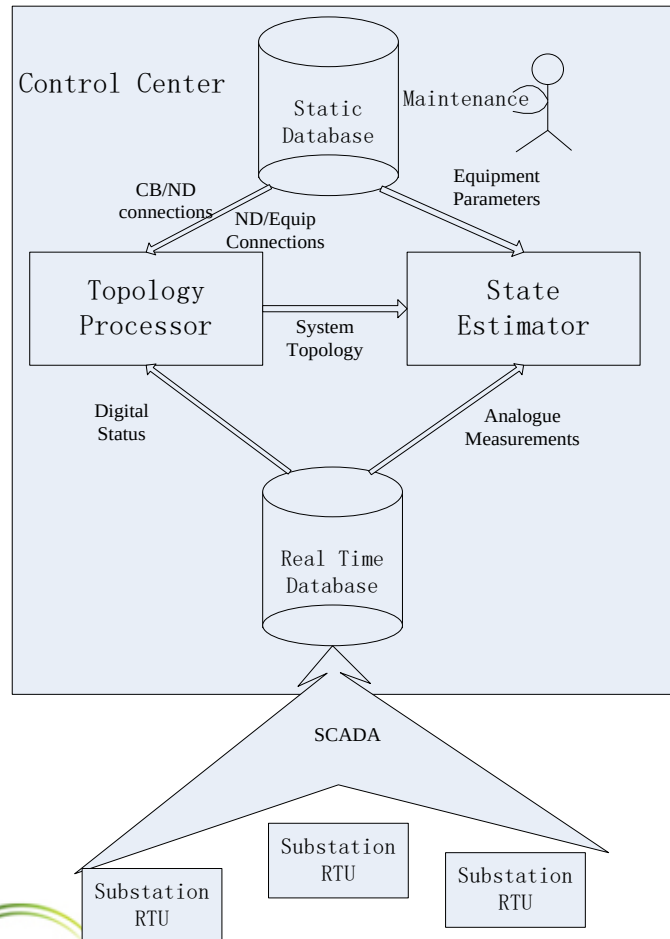
– Solution:

$$\mathbf{x} = (\mathbf{H}^T \mathbf{W} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{W} \mathbf{z}$$

Two-Level Linear State Estimator

- Substation Level
 - Substation Model
 - Circuit Breaker State Estimator
 - Bus Voltage State Estimator
 - Bad Data Detection & Identification
- Control Center Level
 - System Model
 - Topology Processor (system level)
 - State Estimator
 - Bad Data Detection & Identification

Database & Communication Architecture



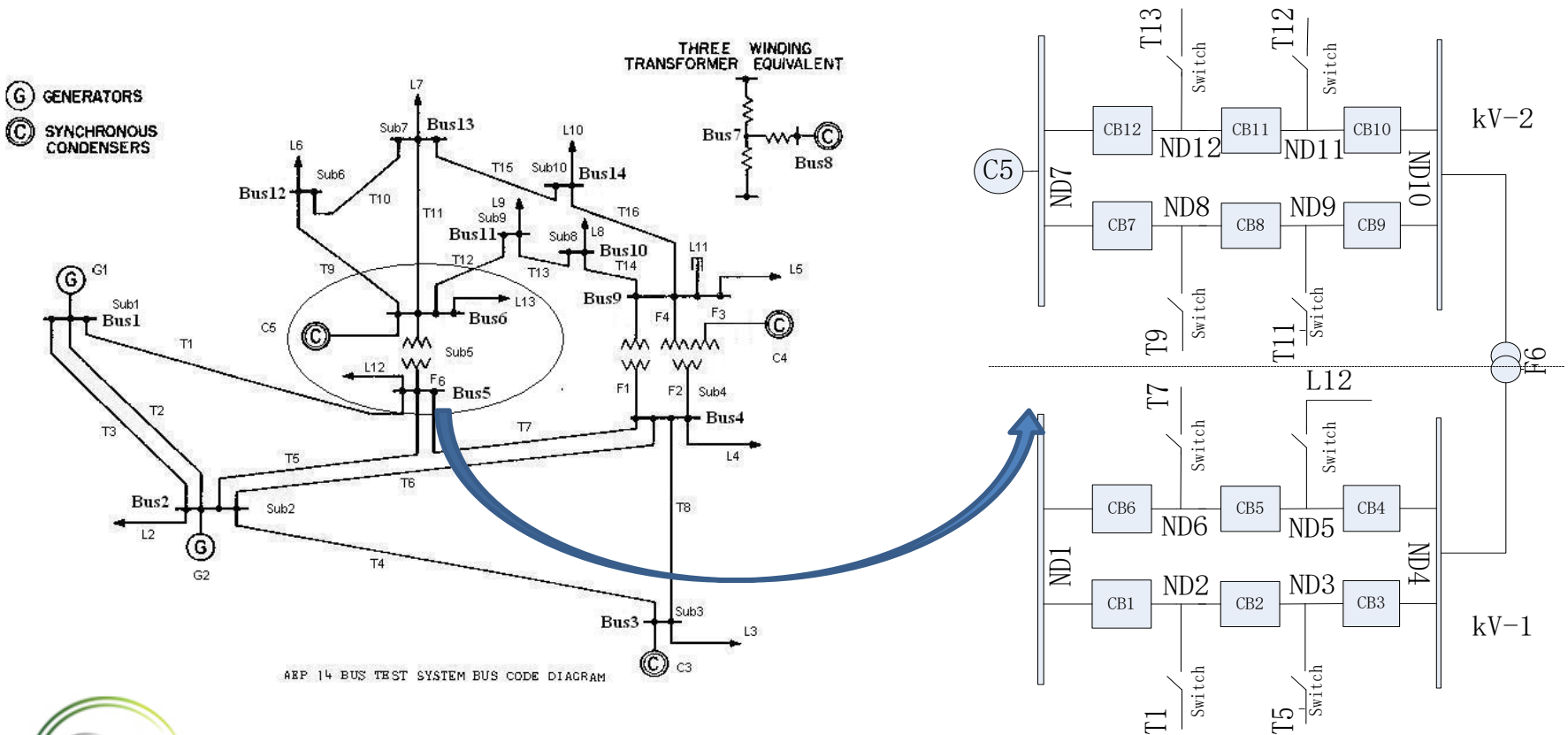
SL-TP: Substation Level Topology Processor

SL-SE: Substation Level State Estimator

RT: Real Time

Substation Level LSE

- Zero-Impedance Current State Estimator
 - Circuit Breaker Oriented Substation Model



Substation Level LSE

– Analog State Estimation

- State:
 - Currents on Circuit Breakers
- Measurements:
 - Injection Currents to Nodes: $\mathbf{z}_{inj}$
 - Currents on Circuit Breakers: $\mathbf{z}_{cb}$
- Measurement Functions
 - Kirchhoff's Current Law
 - Identity Matrix
 - Formula:

$$\mathbf{z} = \begin{pmatrix} \mathbf{z}_{inj} \\ \mathbf{z}_{cb} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{KCL} \\ \mathbf{I} \end{pmatrix} \mathbf{x} + \begin{pmatrix} \mathbf{r}_{inj} \\ \mathbf{r}_{cb} \end{pmatrix} = \mathbf{H}\mathbf{x} + \mathbf{r}$$

Substation Level LSE

- Zero-Impedance Voltage State Estimator

- State:

- Complex Bus Voltages

- Measurements:

- Voltage from PMUs

- Measurement Functions

$$\tilde{\mathbf{z}} = \tilde{\mathbf{H}} \tilde{\mathbf{x}} + \tilde{\mathbf{r}} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \tilde{\mathbf{x}} + \tilde{\mathbf{r}}$$

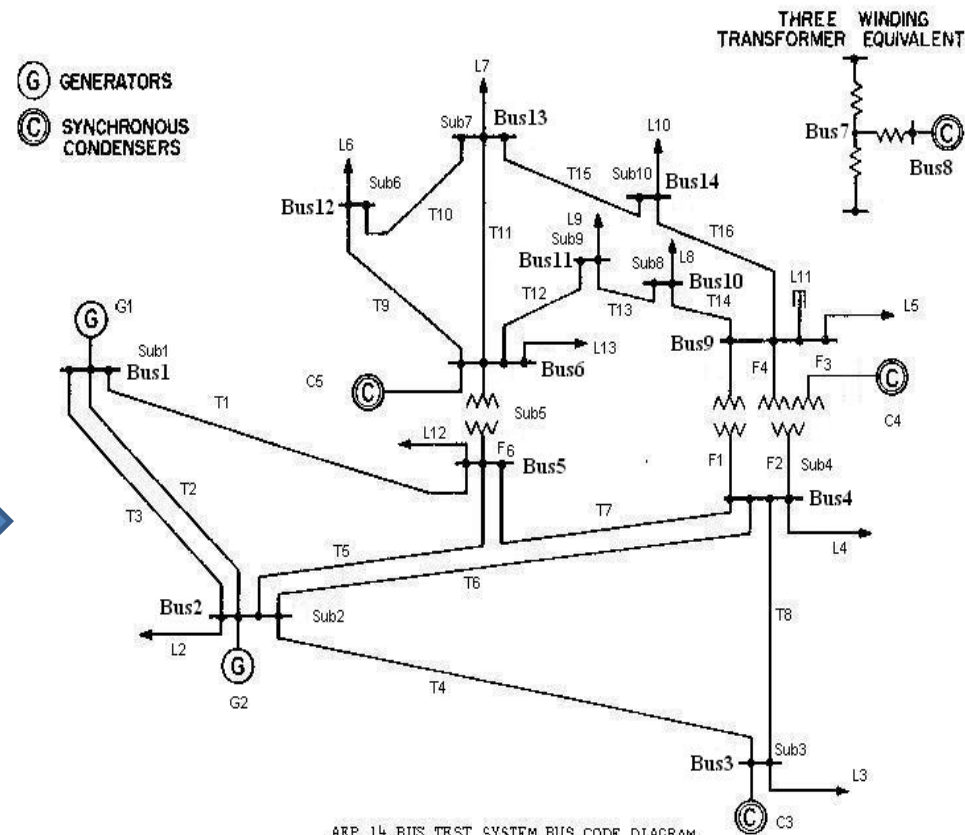
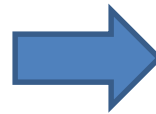
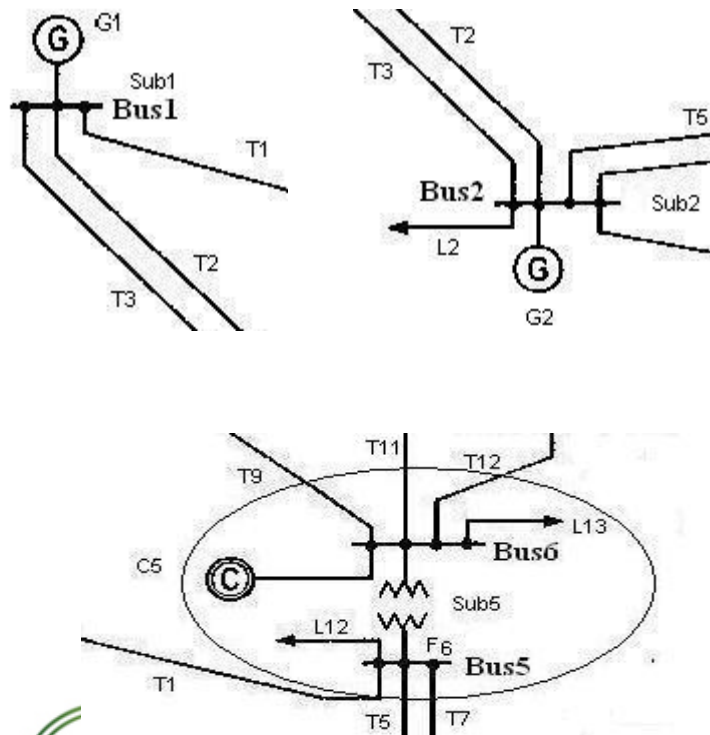
- Solutions:

$$x_{real} = \frac{\sum_{i=1}^m w_{2i-1,2i-1} z_{i,real}}{\sum_{i=1}^m w_{2i-1,2i-1}}$$

$$x_{imag} = \frac{\sum_{i=1}^m w_{2i,2i} z_{i,imag}}{\sum_{i=1}^m w_{2i,2i}}$$

Control Center Level LSE

- Topology Processor
 - Merging Substation Topologies (STDFs)



Control Center Level LSE

- State Estimation

- States

- Complex Bus Voltages

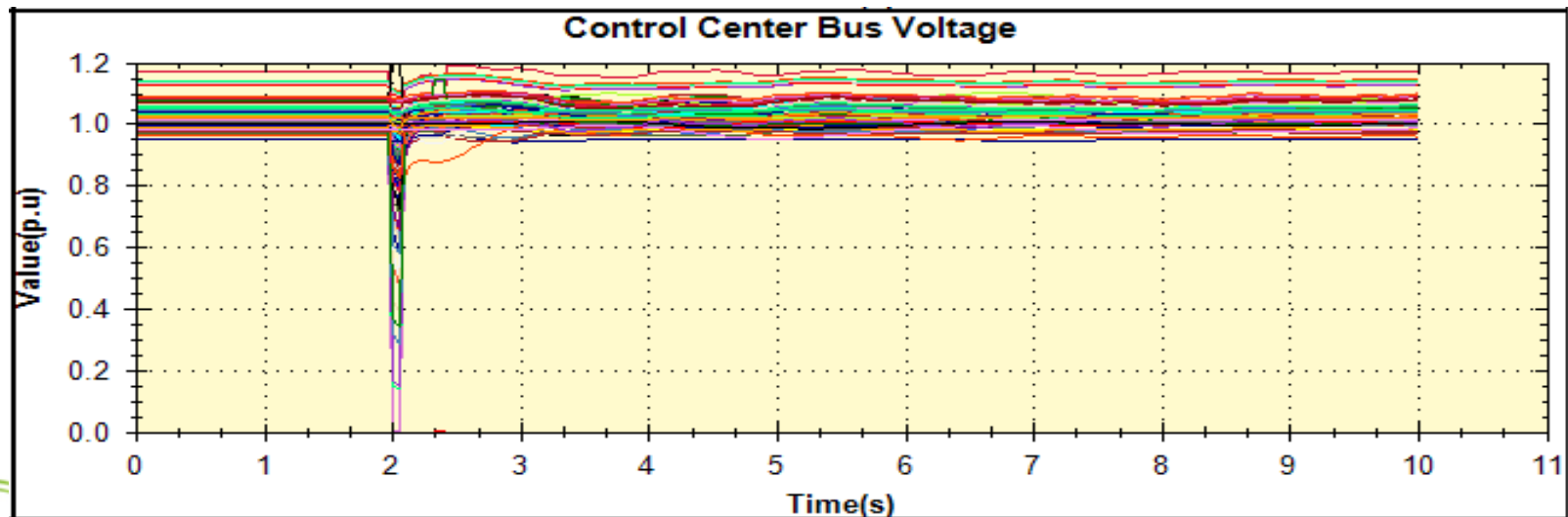
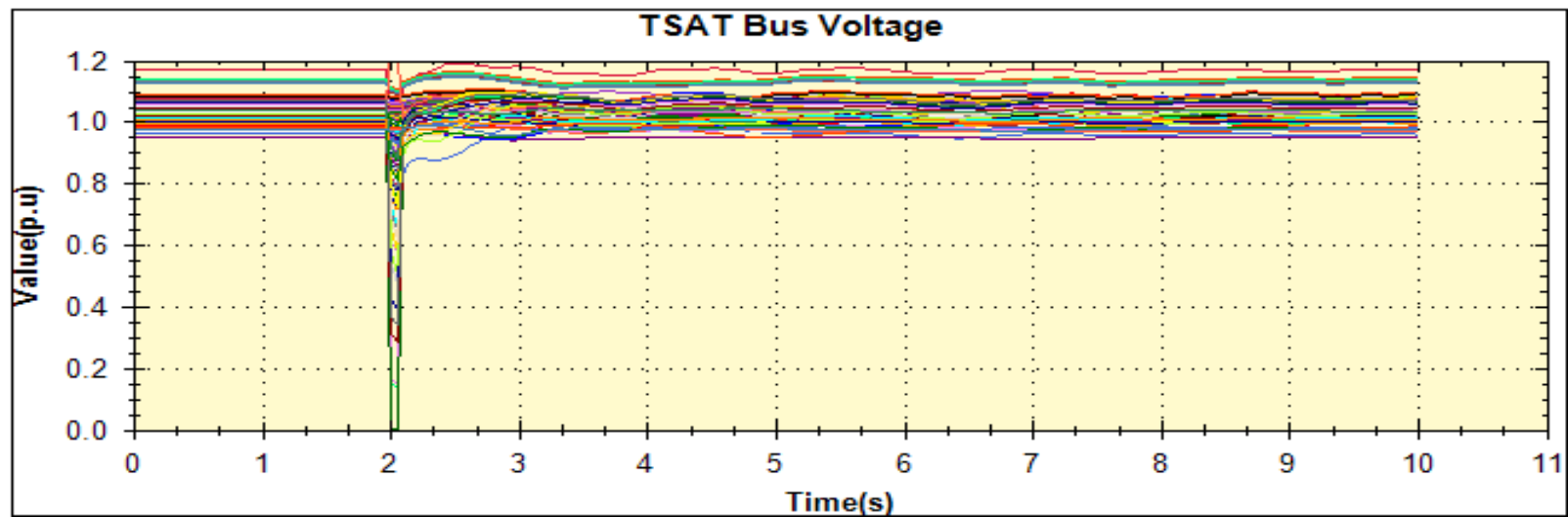
- Measurements (Phasor)

- Bus Voltages $\tilde{\mathbf{V}}_{bus}$
 - Two Direction Branch Currents $\tilde{\mathbf{I}}_{b1}$ $\tilde{\mathbf{I}}_{b2}$
 - Injection Currents $\tilde{\mathbf{I}}_{inj}$

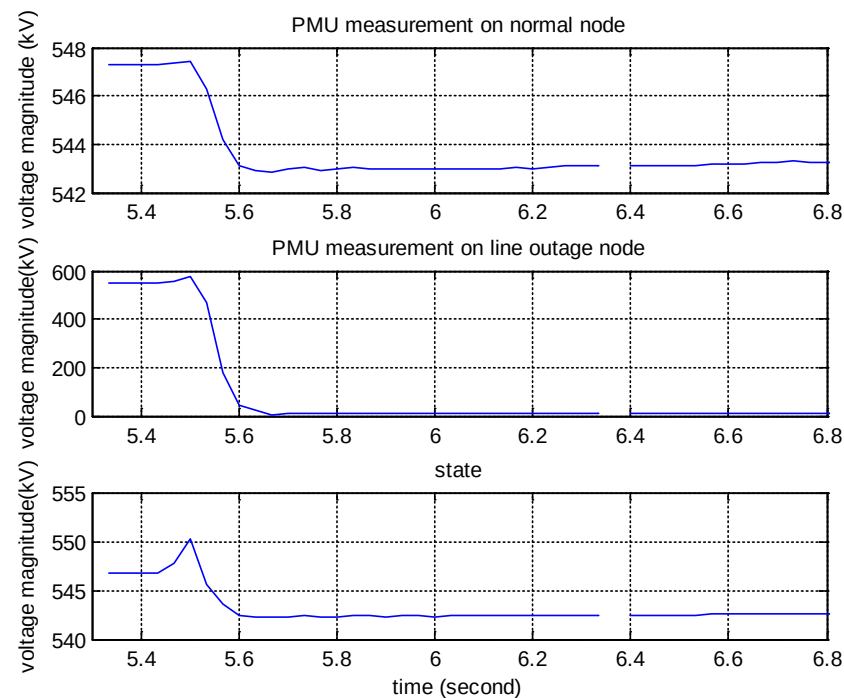
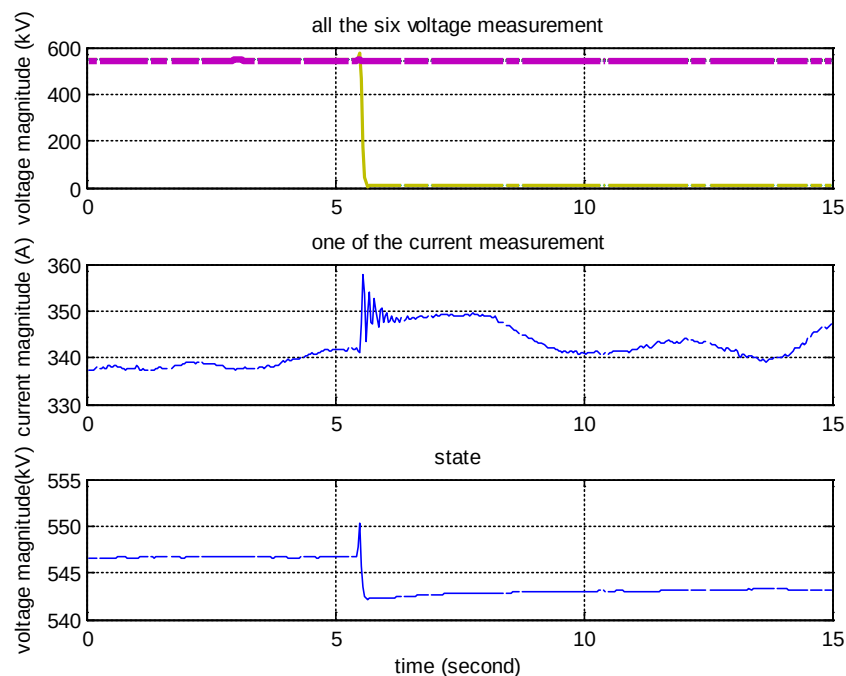
- Measurement Functions

$$\tilde{\mathbf{z}} = \begin{pmatrix} \tilde{\mathbf{V}}_{bus} \\ \tilde{\mathbf{I}}_{b1} \\ \tilde{\mathbf{I}}_{b2} \\ \tilde{\mathbf{I}}_{inj} \end{pmatrix} = \tilde{\mathbf{H}}\tilde{\mathbf{x}} + \tilde{\mathbf{r}} = \begin{pmatrix} \mathbf{I} \\ \tilde{\mathbf{Y}}_1^b \\ \tilde{\mathbf{Y}}_2^b \\ \tilde{\mathbf{Y}} \end{pmatrix} \tilde{\mathbf{x}} + \tilde{\mathbf{r}}$$

WECC 179-Bus System (simulation)



500 kV 10-Sub System (real data)



Uses of LSE

- Like traditional SE, the LSE output can be used as input to downstream EMS applications:
 - Contingency analysis, optimal power flow, etc.
- Unlike traditional SE, LSE can be run 30-60 times a second (at PMU frequencies):
 - LSE output is a filtered (more accurate) version of the PMU measurements; can be used instead of raw measurements
 - LSE output can be displayed to the operator for better situational awareness (especially dynamics)
 - LSE output can be input to SPS or fast controls

Uses of Distributed LSE

- Many applications that can use LSE as input can be run more efficiently at the subs or small regions:
 - Oscillation detection can be run at subs with only the results sent to the EMS (next presentation by Mani V.)
 - Most SPS run with input from one or few subs
 - Automatic voltage control (AVC) is zonal
 - The European AVC (France and others) uses fixed zones
 - The AVC used in most of China (by Sun and Zhang) uses adaptive zones and has put China far ahead of the world in grid voltage control. This AVC has been installed at PJM for the first system-wide voltage control in North America.